

1

Question (2): Indicate the True and the False statements

1. A truly intelligent system adapts itself to deal with changes in problem domain
2. Soft computing can be tolerant of imprecise, incomplete or corrupt input data
3. Conventional computing support learning the solution through repeated observation and adaptation
4. Expert systems are highly specific in a wide band of knowledge
5. Fuzzy logic proposition is either true or false
6. CBR systems solve problems by making use of knowledge about similar problems encountered in the past
7. Techniques such as artificial neural networks and decision trees have made it possible to perform data mining involving large volumes of data
8. Intelligent Software Agents are capable of learning from other intelligent software agents only
9. Language Technology based systems can be the front-end of information systems themselves based on other intelligence tools
10. Factual knowledge is about practice, accurate judgment, and the ability of evaluation, and guessing

Best Wishes

1

11. The major bottleneck in applying Expert Systems technology to new domains is
- (a) Knowledge acquisition
 - (b) Expert system design
 - (c) Expert system development
 - (d) Expert system testing
12. Uncertainty in knowledge domain may include
- (a) Complete and precise knowledge
 - (b) Vague knowledge
 - (c) knowledge introduced by single expert
 - (d) None of the above
13. Which of the following examples can be applied using backward chaining?
- (a) Prediction of share market status
 - (b) Credit assessment
 - (c) Diagnosis of blood cancer in humans
 - (d) Project viability
14. Backward chaining is implemented to answer of the question:
.....
- (a) When this happened?
 - (b) Where this happened?
 - (c) What can happen next?
 - (d) Why this happened?
15. In Forward chaining, all the facts and rules, and sorts them
concluding to a solution
- (a) before
 - (b) during
 - (c) while
 - (d) after

5. In Expert System, the uses rules in knowledge base to arrive at final conclusion
- (a) Knowledge base
 - (b) User Interface
 - (c) Inference Engine
 - (d) Explanation Facility
6.allow us to express knowledge in vague linguistic terms
- (a) Expert Systems
 - (b) Fuzzy Logic
 - (c) ANN
 - (d) Data Mining
7. Genetic Algorithms obtain solutions through a process consisting of
- (a) Survival of the fittest
 - (b) Crossbreeding
 - (c) Mutation
 - (d) All of the above
8. learn from experience and repeated presentation of problems with their corresponding solutions
- (a) Expert Systems
 - (b) ANNs
 - (c) Genetic Algorithms
 - (d) Data Mining
9. Which of the following statements is not considered as one of the characteristics of Soft Computing?
- (a) Solve problems with explicit solution steps
 - (b) Learn the solution through repeated observation and adaptation
 - (c) Can handle information expressed in vague linguistic terms
 - (d) Arrive at an acceptable solution through evolution
10. Which of the following statements is not considered as one of the characteristics of Intelligent System?
- (a) Human like reasoning process
 - (b) Learning from experience
 - (c) Dealing with precise expressions of facts
 - (d) Finding solutions through processes similar to natural evolution

Model A



Faculty of Commerce	كلية التجارة
Information Systems and Computers Department	قسم نظم المعلومات والحاسبات الآلية
Bsc. Program in Management Information Systems	Mid Term Exam, March 2019
Intelligent Decision Support Systems	4 th Year (English+ Arabic) Duration: 45 Minutes

Question 1. Answer the following multiple choice questions

1. Maintenance and extension of a rule base can be difficult for a relatively large rule base, which might exceedsrule(s).
(a) 10000
(b) 1000
(c) 100
(d) 10
2. In test strategy, Expert Systems and a human expert are subjected to blind user evaluation
(a) Field
(b) Expert system against experts
(c) Historical data usage
(d) Turing
3. are usually a system analyst with background in AI who is responsible for choosing the proper knowledge schema
(a) Domain experts
(b) Knowledge engineers
(c) System developers
(d) System users
(e) Turing
4. The following rule: **IF Y is true, conclude: X is true** represent logic.
(a) Modus ponens
(b) Hypothetical syllogism
(c) Modus tollens
(d) Boolean logic

21. Depth-first search always expands the _____ node in the current fringe of the search tree
- Shallowest
 - Child node
 - Deepest
 - Minimum cost
22. Which of the following machine requires input from the humans but can interpret the output themselves?
- Actuators
 - Sensor
 - Agents
 - AI system
23. Which of the following can improve the performance of an AI agent?
- Perceiving
 - Learning
 - Observing
 - All of the mentioned
24. Which of the following is/are the composition for AI agents?
- Program only
 - Architecture only
 - Both Program and Architecture
 - None of the mentioned
25. On which of the following approach A basic line following robot is based?
- Applied approach
 - Weak approach
 - Strong approach
 - Cognitive approach
26. Which of the following is an example of artificial intelligent agent/agents?
- Autonomous Spacecraft
 - Human
 - Robot
 - All of the mentioned
27. What is an AI 'agent'?
- Takes input from the surroundings and uses its intelligence and performs the desired operations
 - An embedded program controlling line following robot
 - Perceives its environment through sensors and acting upon that environment through actuators
 - All of the mentioned
28. Which of the following environment is strategic?
- Rational
 - Deterministic
 - Partial
 - Stochastic
29. In Prolog, which operator is used to represent logical OR?
- &&
 - ,
 - !
 - ;
30. In Prolog, which symbol is used to represent the end of a query?
- !
 - .
 - :
 - ?
31. Which search strategy is also called as blind search?
- Uninformed search
 - Informed search
 - Simple reflex search
 - All of the mentioned
32. Which search is implemented with an empty first-in-first-out queue?

- c) To explain various sorts of intelligence
- d) To solve real-world problems

Which of the following is an application of Artificial Intelligence?

- a) It helps to exploit vulnerabilities to secure the firm
- b) Language understanding and problem-solving (Text analytics and NLP)
- c) Easy to create a website
- d) It helps to deploy applications on the cloud

Breadth-first search always expands the _____ node in the current fringe of the search tree.

- a) Shallowest
- b) Child node
- c) Deepest
- d) Minimum cost

Which of the following is a component of Artificial Intelligence?

- a) Learning
- b) Training
- c) Designing
- d) Puzzling

Prolog is a programming language based on which paradigm?

- a) Imperative programming
- b) Object-oriented programming
- c) Functional programming
- d) Logic programming

In Prolog, knowledge is represented using:

- a) Variables
- b) Loops
- c) Facts and rules
- d) Functions

What is the correct syntax for a Prolog rule representing "X is the mother of Y"?

- a) mother(X, Y) :-
 - b) X(mother, Y) :-
 - c) Y is mother(X) :-
 - d) mother(X) = Y :-
- likes(joe, fish). likes(joe, mary).
likes(mary, fish). likes(albert, joe).

?- likes (Who, fish).

- A) Who = joe.
- B) Who = mary.
- C) Who = albert.
- D) Who = joe. ; Who = mary.

What is the function of an Artificial Intelligence "Agent"?

- Mapping of goal sequence to an action
- Work without the direct interference of the people
- Mapping of precept sequence to an action
- Mapping of environment sequence to an action

depth-first search

b) Breadth-first search

bidirectional search

d) None of the mentioned

Which search algorithm imposes a fixed depth limit on nodes?

depth-limited search

b) Depth-first search

iterative deepening search

d) Bidirectional search

Good Luck

Abeer Amer

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- c) Problem you design d) Representing your problem with variable and parameter

5. What is the main task of a problem-solving agent?

- a) Solve the given problem and reach to goal
- b) To find out which sequence of action will get it to the goal state
- c) All of the mentioned
- d) None of the mentioned

6. What is the full form of "AI"?

- a) Artificially Intelligent
- b) Artificial Intelligence
- c) Artificially Intelligence
- d) Advanced Intelligence

7. Which of the following is/are Uninformed Search technique/techniques?

- a) Breadth First Search (BFS)
- b) Depth First Search (DFS)
- c) Bidirectional Search
- d) All of the mentioned

8. What is Artificial Intelligence?

- a) Artificial Intelligence is a field that aims to make humans more intelligent
- b) Artificial Intelligence is a field that aims to improve the security
- c) Artificial Intelligence is a field that aims to develop intelligent machines
- d) Artificial Intelligence is a field that aims to mine the data

9. Who is the inventor of Artificial Intelligence?

- a) Geoffrey Hinton
- b) Andrew Ng
- c) John McCarthy
- d) Jürgen Schmidhuber

10. Which search implements stack operation for searching the states?

- a) Depth-limited search
- b) Depth-first search
- c) Breadth-first search
- d) None of the mentioned

11. Which of the following is the branch of Artificial Intelligence?

- a) Machine Learning
- b) Cyber forensics
- c) Full-Stack Developer
- d) Network Design

12. What is the goal of Artificial Intelligence?

- a) To solve artificial problems
- b) To extract scientific causes

FACULTY OF BUSINESS
(ENGLISH + ARABIC SECTION) (MODEL.4)

Department : Management Information Systems

Year : 4th Year

Course Title : Artificial Intelligence

Exam : MidTerm

Lecturer : Dr. Abeer A. Amer

Semester : Second

Date : 25/03/2024

Time Allowed : One Hour

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Question .1. True / False:

1. The game of Poker is a single agent.
2. A solution to a problem is a path from the initial state to a goal state. Solution quality is measured by the path cost function, and an optimal solution has the highest path cost among all solutions.
3. likes(jin, jax). the same as likes(jax, jin).
4. food(b, c, d(a)). is ?-food(X, Y). a true query?
5. If sky is blue, everyone likes it. Write this statement as likes(sky, everyone) :- blue(sky).
6. ?- X is -(5,3,1). return an error.
7. ?- 3+2 is 5. same as ?- 5 is 3+2.

Question.2. MCQ

1. Satellite Image Analysis System is (Choose the one that is not applicable).
a) Episodic b) Semi-Static
c) Single agent ☒ d) Partially Observable
2. Categorize Crossword puzzle in Fully Observable / Partially Observable.
☒ a) Fully Observable b) partially Observable
c) All of the mentioned d) None of the mentioned
3. Which were built in such a way that humans had to supply the inputs and interpret the outputs?
a) Agents ☒ b) AI system c) Sensor d) Actuators
4. What is state space?
a) The whole problem b) Your Definition to a problem

11. Cognition deals with trivial software systems that directly interact and communicate with human users.	
12. ANN are sequential computational entities that consist of multiple simple processing units that are connected in specific ways in order to perform the desired tasks.	
13. Weak AI is a set of computer programs that produce output that would be considered to reflect intelligence if it were generated by humans.	
14. If a robot can alter its own trajectory in response to external conditions, it is considered to be intelligent.	
15. Knowledge engineering is concerned with the acquisition, management, use and transformation of knowledge.	
16. The two big tasks of CBR are:- Classification tasks and Synthesis tasks.	
17. Knowledge and reasoning also play a crucial role in dealing with Completely Observable environment.	
18-Treatment chosen by doctor for a patient for a disease is based on only current symptoms.	
19-CBR solves new problems by adapting old solutions to similar problems. It offers several potential advantages over rule-based reasoning: rules are not combined blindly in a search for solutions, solutions can be explained in terms of concrete examples, and performance can Improve automatically as new problems are solved and added to the case library.	
20-Case-based reasoning has been formalized for purposes of computer reasoning as a four-step process: Retrieve, Reuse, Revise, and Retain,	

Question 2. (15 Minutes, 5 Marks)

Explain in details the following terms:-

1-The "Chinese room" experiment proceeds.

2-Linguistic Level in Brachman's 5 Levels of Knowledge.

3-Automatic theorem proving.

4-CBR System Components.

5-Advantages & Disadvantages of CBR.

Best Wishes

Faculty of Commerce		كلية التجارة	
Information Systems and Computers Department		قسم نظم المعلومات والحاسبات الآلية	
Bsc. Program in MIS		Dr. Mahmoud Farouk	Date: 5 May 2021
Intelligent decision support system	4 th Year (English)	Time: 45 Minutes	

Question 1: True or False (Correct the false answer)
(30 Minutes, 10 Marks)

Statement	Answer
1. Intelligence denotes the ability of an individual to adapt his thinking to new demands; it is the common mental adaptability to new tasks and conditions of life.	
2. In "Chinese Room" Searle conclude that 'manipulating formal symbols, which is just what a computer running a program does, is sufficient for understanding or thinking'.	
3. An AI agent perceives and acts upon the environment using sensors and perceiver.	
4. A major thrust of AI is in the development of computer functions associated with human intelligence.	
5. A computer program with AI can not answer the generic questions it is meant to solve.	
6. Generic inference pattern "Heuristic Classification" describes the problem-solving behavior of these systems on the Knowledge Level in a generic way.	
7. An expert system is a software application that stores knowledge about multi domains. It is able to draw conclusions from that knowledge and offers concrete solutions for problems in that domains.	
8. Analytical term describes the common-sense of the judgmental part of problem-solving.	
9. The propose & revise method is an efficient method for solving the task of parametric design. It depends on the following inferences:- propose, C-test, and revise.	
10. Deep Blue was chess-playing computer developed by IBM that won against world champion Garry Kasparov in 1997. It was fine-tuned by chess grand masters.	

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Bsc. Program in MIS		Dr. Mahmoud Farouk	Date: 2 Abril 2022
Intelligent decision support system	4 th Year (English + Arabic)		Time: 45 Minutes

Question 1: Explain in details the following sentences
(30 Minutes, 10 Marks)

Sentence	Marks
manipulating formal symbols, which is just what a computer running program does, is not sufficient for understanding or thinking (Chinese room).	2
The difference between Strong AI & Weak AI .	2
Newell's 3 levels of knowledge.	1
Image Understanding (IU) deals with the analysis and interpretation of visual information.	1
CBR System Components.	1
CBR Knowledge Containers.	1
Matching strings in CBR (Matching strings, spelling check, word-count).	1
How to Adapt the Solution in CBR?	1

Question 2. Define the following terms
(15 Minutes, 5 Marks)

a-Deduction.	b-Induction.	c-Blind Search.	d-Expert Systems.	e- Planning.
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